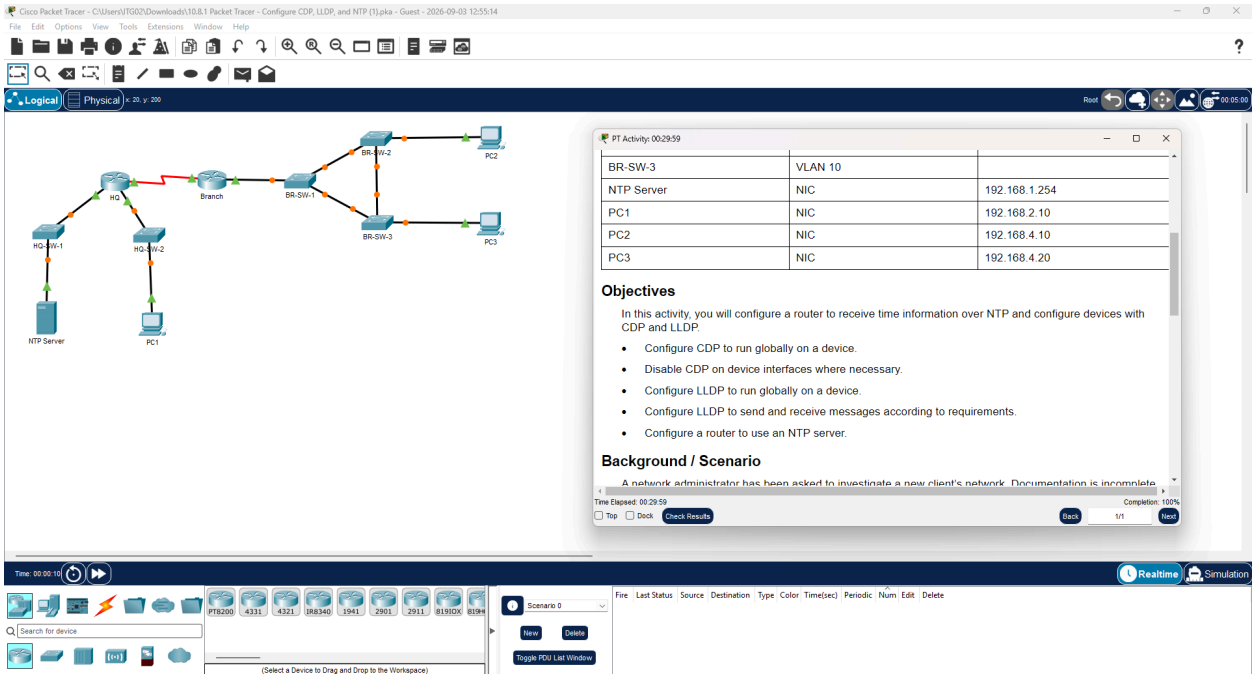


1. Executive Summary & Objectives

This laboratory report details the technical implementation, network discovery, device configuration, and security hardening performed for Packet Tracer Activity 10.8.1. The primary focus of this exercise is to configure Cisco Discovery Protocol (CDP), Link Layer Discovery Protocol (LLDP), and Network Time Protocol (NTP) across a multi-site network architecture consisting of HQ and Branch locations.

Key Objectives

- **CDP Management:** Globally enable CDP on designated Branch devices while selectively disabling CDP on untrusted access interfaces to prevent network reconnaissance.
- **LLDP Hardening:** Disable CDP and enable vendor-neutral LLDP on HQ site devices; fine-tune interface directionality (Transmit vs. Receive) to comply with security guidelines.
- **Network Discovery:** Utilize CDP neighbor commands over SSH to dynamically discover unknown IP management addresses of remote switches.
- **Time Synchronization:** Configure Network Time Protocol (NTP) on the primary router to ensure precise, synchronized logging across network infrastructure.

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2. Network Topology & Architecture Overview

The topology connects two distinct operational sites over a point-to-point serial link:

```
None
                          [ NTP Server ]
                             (192.168.1.254)
                                 | F0/24
                                 |
[ HQ-SW-1 ] <--- G0/1 ---> [ HQ Router ] <--- S0/1/0 ---> [ Branch Router ] <---
G0/0/0 ---> [ BR-SW-1 ]

                                | G0/0/1                        (192.168.3.2/30)
|
                                | G0/1
|-- BR-SW-2 (F0/1 -> PC2)
                          [ HQ-SW-2 ] <--- F0/1 ---> [ PC1 ]
|-- BR-SW-3 (F0/1 -> PC3)
```

1. **HQ Site (LLDP & NTP Domain):** Uses vendor-neutral LLDP. The HQ router acts as an NTP client synchronizing with an authoritative NTP server (`192.168.1.254`). Switches `HQ-SW-1` and `HQ-SW-2` uplink to HQ via Gigabit ports.
2. **Branch Site (CDP & Management Domain):** Uses Cisco proprietary CDP. The Branch router connects via Serial to HQ and via Ethernet to switch `BR-SW-1`. Switches `BR-SW-2` and `BR-SW-3` are managed remotely via SSH through `BR-SW-1`.

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3. Complete Addressing Table

The table below reflects all network interfaces, assigned IP parameters, and management addresses discovered during the execution of this lab.

Device	Interface	IP Address	Subnet Mask / Prefix	Notes / Connection Details
HQ Router	G0/0/0	192.168.1.1	255.255.255.0 (/24)	Uplink to ...
HQ Router	G0/0/1	192.168.2.1	255.255.255.0 (/24)	Uplink to ...
HQ Router	S0/1/0	192.168.3.1	255.255.255.252 (/30)	WAN Poi...
Branch R...	G0/0/0	192.168.4.1	255.255.255.0 (/24)	Gateway ...
Branch R...	S0/1/0	192.168.3.2	255.255.255.252 (/30)	WAN Poi...
HQ-SW-1	VLAN 1	Not Configured	N/A	Layer 2 S...
HQ-SW-2	VLAN 1	Not Configured	N/A	Layer 2 S...
BR-SW-1	VLAN 10	192.168.4.250	255.255.255.0 (/24)	Discover...
BR-SW-2	VLAN 10	192.168.4.253	255.255.255.0 (/24)	Discover...
BR-SW-3	VLAN 10	192.168.4.254	255.255.255.0 (/24)	Discover...
NTP Server	NIC	192.168.1.254	255.255.255.0 (/24)	Dedicate...
PC1	NIC	192.168.2.10	255.255.255.0 (/24)	Host con...

Device	Interface	IP Address	Subnet Mask / Prefix	Notes / Connection Details
PC2	NIC	192.168.4.10	255.255.255.0 (/24)	Host con…
PC3	NIC	192.168.4.20	255.255.255.0 (/24)	Host con…

---

4. Complete Device Configurations

4.1. HQ Site Configuration (LLDP & NTP)

HQ Router CLI Configuration

o
enable
configure terminal
hostname HQ

! Disable Cisco Discovery Protocol globally
no cdp run

! Enable Link Layer Discovery Protocol globally
lldp run

! Configure interfaces connecting to HQ switches to receive LLDP messages only
interface GigabitEthernet0/0/0
description Connection to HQ-SW-1
no lldp transmit
lldp receive
exit

interface GigabitEthernet0/0/1
description Connection to HQ-SW-2
no lldp transmit
lldp receive
exit

! Configure NTP Server Client Synchronization
ntp server 192.168.1.254

end
copy running-config startup-config

None

```
#### HQ-SW-1 Switch CLI Configuration
```cisco
enable
configure terminal
hostname HQ-SW-1

! Disable CDP and enable LLDP globally
no cdp run
lldp run

! Uplink to HQ Router: Send-only LLDP messages
interface GigabitEthernet0/1
 description Uplink to HQ Router G0/0/0
 no lldp receive
 lldp transmit
 exit

! Access Port to NTP Server: Completely disable LLDP
interface FastEthernet0/24
 description Connection to NTP Server
 no lldp receive
 no lldp transmit
 exit

end
copy running-config startup-config
```
```

HQ-SW-2 Switch CLI Configuration

```
None
enable
configure terminal
hostname HQ-SW-2

! Disable CDP and enable LLDP globally
no cdp run
lldp run

! Uplink to HQ Router: Send-only LLDP messages
interface GigabitEthernet0/1
 description Uplink to HQ Router G0/0/1
 no lldp receive
 lldp transmit
 exit

! Access Port to PC1: Completely disable LLDP
interface FastEthernet0/1
 description Connection to PC1
 no lldp receive
 no lldp transmit
 exit

end
copy running-config startup-config
```

4.2. Branch Site Configuration (CDP & Remote SSH Management)

Branch Router CLI Configuration

```
None
enable
configure terminal
hostname Branch

! Activate CDP globally on Branch Router
cdp run
```

```
end
copy running-config startup-config
```

BR-SW-1 Switch Configuration (SSH Session from Branch Router)

```
None
! Connected via SSH from Branch Router: ssh -l admin 192.168.4.250
! Password: SW1admin# / Enable Secret: SW1EnaAccess#

enable
configure terminal
! CDP is enabled by default; used to discover BR-SW-2 and BR-SW-3
end
copy running-config startup-config
```

BR-SW-2 Switch CLI Configuration (SSH Session from BR-SW-1)

```
None
! Connected via SSH from BR-SW-1: ssh -l admin 192.168.4.253
! Password: SW2admin# / Enable Secret: SW2EnaAccess#

enable
configure terminal

! Disable CDP on active user access port connected to PC2
interface FastEthernet0/1
 description Access Port to PC2
 no cdp enable
 exit

end
copy running-config startup-config
```

BR-SW-3 Switch CLI Configuration (SSH Session from BR-SW-1)

```
None
! Connected via SSH from BR-SW-1: ssh -l admin 192.168.4.254
! Password: SW3admin# / Enable Secret: SW3EnaAccess#
```

```
enable
configure terminal

! Disable CDP on active user access port connected to PC3
interface FastEthernet0/1
 description Access Port to PC3
 no cdp enable
 exit

end
copy running-config startup-config
```

---

5. Implementation & Network Discovery Walkthrough

Part 1: HQ Site LLDP Deployment & Security Directionality

1. **Global Configuration:** CDP was globally disabled on `HQ`, `HQ-SW-1`, and `HQ-SW-2` using `no cdp run`. LLDP was globally enabled using `lldp run`.
2. **Interface Direction Adjustment:**
    - On `HQ`, ports `G0/0/0` and `G0/0/1` were set to `no lldp transmit` and `lldp receive`. This allows `HQ` to monitor connected switches without advertising its own routing details downstream.
    - On `HQ-SW-1` and `HQ-SW-2`, uplink interfaces (`G0/1`) were set to `no lldp receive` and `lldp transmit`, completing the unidirectional discovery hierarchy.
3. **Access Port Hardening:** Unused LLDP traffic was blocked on end-device interfaces (`HQ-SW-1` `F0/24` and `HQ-SW-2` `F0/1`) by explicitly issuing `no lldp receive` and `no lldp transmit`.

Part 2: Branch Site CDP Activation & Dynamic Remote Discovery

1. **Branch Router CDP Activation:** Executed `cdp run` on `Branch`.
2. **Discovering BR-SW-1:** Issuing `show cdp neighbors detail` on `Branch` revealed the management IPv4 address of `BR-SW-1` (`192.168.4.250`).

3. **SSH Access to BR-SW-1:** Connected using `ssh -l admin 192.168.4.250` with password `SW1admin#`.
4. **Discovering BR-SW-2 & BR-SW-3:** From `BR-SW-1`, running `show cdp neighbors detail` identified:
    - **BR-SW-2:** Management IP `192.168.4.253`
    - **BR-SW-3:** Management IP `192.168.4.254`
5. **Access Port CDP Disabling:** Standard SSH sessions were established to `BR-SW-2` and `BR-SW-3`. The interface-level command `no cdp enable` was applied on port `FastEthernet0/1` on both switches to stop broadcasting CDP packets toward user endpoints (`PC2` and `PC3`).

Part 3: NTP Synchronization

1. **HQ Configuration:** The command `ntp server 192.168.1.254` was executed on `HQ`.
2. **Convergence:** Allowed time acceleration via Fast Forward Time in Packet Tracer to complete the NTP handshake.

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6. Verification & Output Analysis

6.1. Verification of LLDP on HQ Router

Executing `show lldp neighbors` on `HQ` displays neighbor switches while ensuring no outbound advertisements are leaked on restricted interfaces:

```
None
HQ# show lldp neighbors
Capability codes:
    (R) Router, (B) Bridge, (T) Telephone, (C) Docsis Cable Device
    (W) WLAN Access Point, (P) Repeater, (S) Station, (O) Other

Device ID           Local Intf     Hold-time  Capability      Port ID
HQ-SW-1             Gig0/0/0       120        B               Gig0/1
HQ-SW-2             Gig0/0/0/1     120        B               Gig0/1

Total entries displayed: 2
```

6.2. Verification of CDP Discovery on Branch Router

Running `show cdp neighbors detail` on `Branch` provided the required IP information to begin remote switch administration:

```
None
Branch# show cdp neighbors detail
-------------------------
Device ID: BR-SW-1
Entry address(es):
  IP address: 192.168.4.250
Platform: cisco 2960,  Capabilities: Switch IGMP
Interface: GigabitEthernet0/0/0,  Port ID (outgoing port): FastEthernet0/1
Holdtime : 162 sec
```

6.3. Verification of NTP Status on HQ Router

Verification of NTP clock synchronization with `192.168.1.254`:

```
None
HQ# show ntp associations

address         ref clock       st   when   poll  reach  delay  offset   disp
*~192.168.1.254  127.127.1.1      1     14     64    377  0.211   0.012  0.154
 * sys.peer, # selected, + candidate, - configured

HQ# show ntp status
Clock is synchronized, stratum 2, reference is 192.168.1.254
nominal freq is 250.0000 Hz, actual freq is 249.9990 Hz, precision is 2**24
reference time is E8A12B04.1A2B3C4D (17:03:20.123 UTC Wed Sep 9 2026)
clock offset is 0.0120 msec, root delay is 0.2110 msec
```

---

7. Lab Reflection & Technical Questions

Question 1: Why is it necessary to disable CDP and restrict LLDP on user-facing/access interfaces and untrusted links?

Answer:
CDP and LLDP broadcast unencrypted multicast frames containing sensitive architectural details, including device hostnames, IOS software versions, IP management addresses, hardware platforms, native VLAN IDs, and duplex settings. If enabled on access ports connected to untrusted end-user devices or external links, an attacker running a packet sniffer (such as Wireshark) can easily collect this information. This process—known as passive reconnaissance—allows attackers to map out the network topology and target known vulnerabilities in specific IOS versions. Disabling these protocols on user-facing ports adheres to the principle of Least Privilege and Network Hardening.

Question 2: What information from show cdp neighbors detail was used to SSH into BR-SW-1?

Answer:
The **IPv4 Management Address** (`192.168.4.250`) associated with `BR-SW-1` listed under the `Entry address(es):` section was used to establish the SSH session directly from the Branch router CLI via `ssh -l admin 192.168.4.250`.

Question 3: What were the discovered IP addresses for BR-SW-2 and BR-SW-3?

Answer:

- **BR-SW-2:** `192.168.4.253` (Discovered via `show cdp neighbors detail` on `BR-SW-1`)
- **BR-SW-3:** `192.168.4.254` (Discovered via `show cdp neighbors detail` on `BR-SW-1`)

Question 4: Why was `no cdp enable` used on individual switch interfaces instead of `no cdp run` globally on BR-SW-2 and BR-SW-3?

Answer:
Using `no cdp run` in global configuration mode shuts down the CDP process entirely across the entire switch. Doing so would prevent `BR-SW-2` and `BR-SW-3` from sharing

diagnostic discovery data over their inter-switch trunk links with `BR-SW-1`. In contrast, applying `no cdp enable` under specific interface subconfigurations (e.g., `interface FastEthernet0/1`) selectively stops CDP frame transmission only on those specific access ports connected to host PCs, maintaining internal infrastructure visibility while securing the edge.

Question 5: How do you verify that the HQ router has successfully synchronized its clock with the NTP server?

Answer:
Clock synchronization is verified using two primary commands:

1. `show ntp associations`: A successful synchronization is indicated by an asterisk (`*`) symbol immediately preceding the NTP server IP address (e.g., `*~192.168.1.254`), showing that the router has selected it as the reference system clock source.
2. `show ntp status`: The output explicitly confirms status by displaying `Clock is synchronized, stratum 2, reference is 192.168.1.254`.

---

8. Real-World Best Practices & Recommendations

1. **Protocol Standardization:** Use vendor-neutral LLDP (IEEE 802.1AB) in multi-vendor enterprise environments rather than proprietary CDP to ensure consistent discovery across diverse hardware platforms.
2. **Default Security Stance:** In enterprise deployments, discovery protocols should be disabled globally by default on access-layer switches and enabled only on specific trunk links connecting known, managed infrastructure devices.
3. **NTP Authentication:** Implement NTP MD5 key authentication (`ntp authenticate`, `ntp authentication-key`, and `ntp trusted-key`) in production networks to prevent malicious time spoofing attacks, which can compromise log integrity and cryptographic certificate validation.
4. **Redundant NTP Hierarchy:** Configure at least two distinct NTP sources (Stratum 1 or Stratum 2) on core devices to maintain accurate time distribution in the event of primary server failure or WAN degradation.